

Renesas Device Partition Manager

 Renesas Device Partition Manager

✖ At least one action must be selected

Device Family:

Renesas RA

Action

☐ Read current device information

☐ Change debug state

☐ Set TrustZone secure / non-secure boundaries

☐ Initialize device

Target MCU connection:

J-Link

Connection Type:

SCI

Emulator Connection:

Serial No

Serial No/IP Address:

Debugger supply voltage (V):

0

Connection Speed (bps for SCI, Hz for SWD):

9600

Debug state to change to:

Secure Software Development

Memory partition sizes

☐ Use Renesas Partition Data file

Browse...

Code Secure (KB)

32

Code NSC (KB)

0

Data Start Address (KB)

Data Secure (KB)

0

SRAM Secure (KB)

8

SRAM NSC (KB)

0

Show Command Line

Run



Close

The Renesas Device Partition Manager is used to configure a TrustZone device prior to debugging.

It can perform the following actions on the TrustZone device.

Read current device information

Change debug state

Set TrustZone secure / non-secure boundaries

Initialize device

Selecting an action and clicking on the Run button will execute it on the TrustZone device.

Actions

Read current device information

This action will read the status of the device and display it on the console.

Change debug state

This action will change the debug state of the device, when this action is checked the "Debug state to change to" combo box will be enabled and it is possible to select which state to transition to, the states are:

Secure Software Development

Non-secure Software Development

Non-debugging state

Set TrustZone secure /non-secure boundaries

This action will program the secure/non-secure memory partitions on the device, when this action is checked the "Memory partition sizes" will be enabled and it is possible to enter the values for the memory partitions in KB.

Values (in KB) must be entered for all memory partitions or else it will not be possible to execute the action (the Run button will be disabled).

For partitions that are not supported for a specific device, please input 0 for the value.

Initialize device

This action will initialize the device's flash memory content and memory partition settings to their initial values. When this action is selected all other actions will be disabled.

Target MCU connection

The Target MCU connection combo box will display the available emulators which can be used to connect to the TrustZone Device, these are:

J-Link - Segger J-Link emulator

E2 - Renesas E2 emulator

E2 Lite - Renesas E2 Lite emulator

IAR I-jet (x64) - IAR I-jet emulator; supported for Windows x64 environment only

PEmicro Cyclone FX - PEmicro Cyclone FX emulator

The result of each action executed can be seen in the console area at the bottom.

It is possible to import and export the Device Partition manager settings using the Import and Export buttons.

Running the Renesas Device Partition Manager

The Renesas Device Partition Manager can be run from the **Run** → **Renesas Debug Tools** → **Renesas Device Partition Manager** menu item in RASC and e2 studio workbench.

The Renesas Device Partition Manager can be run as a standalone application from the RASC command line as follows:

```
rasc.exe -application com.renesas.cdt.ddsc.dpm.ui.dpmapplication
```

The following arguments can be passed to the standalone application to initialise the Renesas Device Partition Manager dialog with the memory partitions of a project:

1. The location of the configuration.xml of a project
2. The location of the elf file of a project

E.g. `rasc.exe -application com.renesas.cdt.ddsc.dpm.ui.dpmapplication <PATH_TO_PROJECT>\configuration.xml <PATH_TO_PROJECT>\Debug\RA6M4.elf`

Reading the memory partitions from a Renesas Partition Data File

There is a field above the Memory partition sizes to allow the Renesas Device Partition Manager dialog to be initialised with the memory partitions from a specified Renesas Partition Data (.rpd) file. To specify the rpd file to read the memory partitions from, check the Use Renesas Partition File checkbox and enter the path to the rpd file in the field below the checkbox or select the rpd file using the Browse... button.

The Renesas Device Partition Manager can be run from the following 3rd party tools

IAR EWARM - The Renesas Device Partition Manager can be launched from the **Tools** → **Configure Tools** menu item

Configure Tools

Menu Content:

DPM

Menu Text:

DPM

Command:

C:\Users\A5100612\AppData\Local\Programs\Renesas\fsp_sc\202304\2023-04.120230227-1237\eclipse\rasc.exe

Argument:

-application com.renesas.cdt.ddsc.dpm.ui.dpmapplication

Initial Directory:

☐ Redirect to **O**utput Window

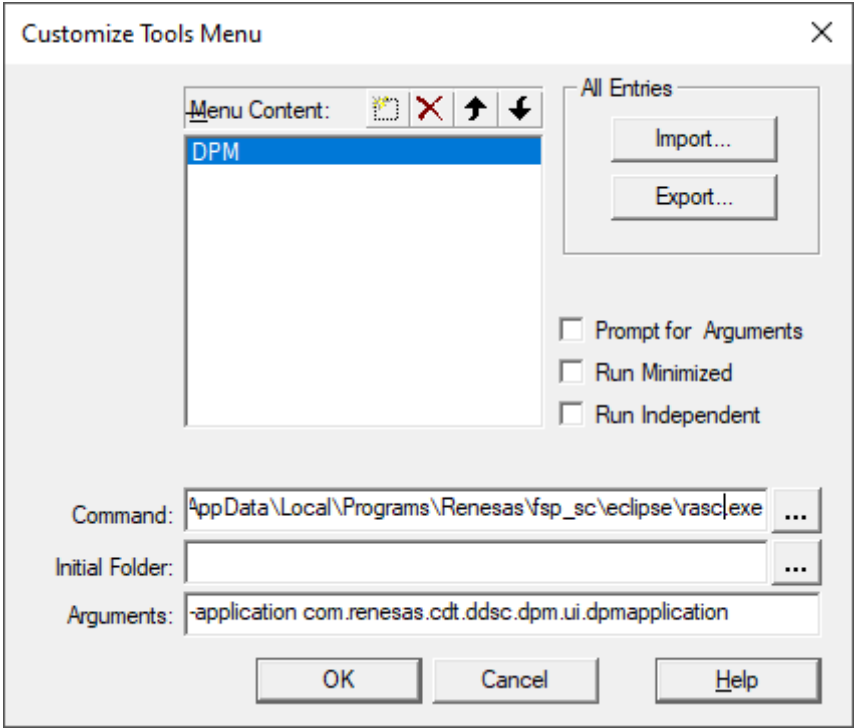
☐ Prompt for Command Line

Tool Available:

Always

Buttons: OK, Cancel, New, Delete, Browse...

KEIL MDK - The Renesas Device Partition Manager can be launched from the **Tools** → **Customize Tools Menu...** menu item



The Renesas Device Partition Manager can be run as a command line interface (cli), the command line interface is located in the following location in e2 studio support files:

<SUPPORT_FILE_LOCATION>\DebugComp\RA\DevicePartitionManager\RenesasDevicePartitionManagerCmd.exe

(the value of SUPPORT_FILE_LOCATION can be found from **Help** → **About** → **Installation Details** → **e2 studio support area** in e2 studio or RASC)

Commands

- Each command must have a hyphen "-" as the prefix.
- All commands and their respective arguments are NOT case sensitive.

Command Name: -action

Prerequisite: Must be included in all cases.

Description: The main action chosen by the user.

A combination of actions can be used to carry out a series of operations in one execution. Rule of combination: If there are multiple arguments, each -action argument must be followed by a ",". The last argument in the -action command must not have ",", else error will be returned and program will be exited.

Arguments:

STATUS	Read device’s current status and display on the console
DLM	Transition DLM state. Requires the -dlmTargetState command.

BOUNDARY	Program IDAU memory boundaries. Must be used together with at least one of the -idau commands.
INITIALIZE	Initialize the device (erase all). Initialization of device cannot happen when the current state of the device is in CM. Note: this is a standalone action as any other actions will be ignored when used together.
PL	Transition the Protection Level (PL) to a target PL. Must be used together with the command -plTarget.

E.g.: -action DLM or -action dlm

Combination of commands: (regardless of order)

-action DLM, BOUNDARY

Program memory boundaries. Move target state to NSECSD from SSD.

-action STATUS, DLM

Transition from one debug state to another. Display current status of device (Debug state and boundaries)

-action STATUS, BOUNDARY

Program memory boundaries. Display current status of device (Debug state and boundaries)

-action STATUS, DLM, BOUNDARY

Program memory boundaries. Transition from one debug state to another. Display current status of device (Debug state and boundaries)

-action DLM, STATUS, BOUNDARY, INITIALIZE

Any other actions when used together with INITIALIZE will be completely ignored. Will be treated as -action INITIALIZE.

Command Name: -deviceFamily

Prerequisite: Optional. If omitted, RA family is selected by default.

Description: Specify the device family of the device.

Arguments:

RA E.g.: -deviceFamily RA

Command Name: -emuType

Prerequisite: Must be included in all cases.

Description: Emulator type being used.

Can be E2, E2 Lite, J-Link or other debug probes.

Arguments:

E2 - Renesas E2 emulator

E2LITE - Renesas E2 Lite emulator

JLINK - Segger J-Link emulator

OTHER - Emulator other than the above. Used together with -vendorProbe command. E.g.: -emuType E2

Command Name: -bootInterface

Prerequisite: Optional.

By default "SCI" is selected as the boot mode connection interface.

Description: Connection interface for boot mode.

Arguments:

SCI - SCI (2-wired UART) connection interface

SWD - SWD connection interface E.g: -bootInterface SWD

Command Name: -emuConn

Prerequisite: Optional.

By default it is chosen as "serial" which is serial number.

Description: This is only supported for J-Link where user can select the connection method to specify an emulator either through "serial number", "select IP by serial number" or "IP address".

Arguments: For E2/E2 Lite, the connection method is strictly emulator serial number:

serial - Serial number

serialip - Select an emulator which is connected to the host via Ethernet, by its serial number.

ipaddr - IP address E.g.: -emuConn ipaddr

Command Name: -emuName

Prerequisite: Optional.

Description: The input for this depends on what is being chosen by -emuConn.

If -emuConn is omitted, serial number of emulator is used to identify the emulator to connect to.

For J-Link, "select IP by serial number" or "IP address" method can be used to connect to the emulator depending on the method chosen using the "-emuConn" command.

Optional: Command line tool will automatically choose the first readily available emulator to connect to if this command is not included.

Note: J-Link supports this from 1.2.1.0 alpha version and onwards.

Arguments: Serial number of the emulator - Specify debugging with the emulator chosen.

E.g.: -emuName 4IM000007

Command Name: -supplyVoltage

Prerequisite: None.

Optional.

Description: Set the power supply (in V) to the device.

Optional: For E2/E2 Lite, omitting this command will automatically set the supply voltage to 0, which means external power will be needed. For J-Link, this command is unnecessary.

Arguments: Voltage of the device in numeric value.

Note: The input value has to be 100 times the value with 2 decimal points. 3.3V would require an input of "330". Please set the value to 0 if you wish to not supply any voltage through the emulator.

E.g: -supplyVoltage 330

Command Name: -connBaudRate (SCI boot mode)

Prerequisite: None.

Optional.

Description: The baud rate to be used during the connection. (in bps)

Optional: Omitting this command will automatically set the baud rate to the default baud rate specified for E2/E2 Lite. J-Link is fixed to 9600bps regardless of the argument.

Note: The value will be adjusted to the smaller supported value if unsupported value is entered. If the entered value is below the minimum requirement, it'll be adjusted to the minimum value of 9600.

Arguments: Baud rate in numeric value of bps. Below is the list of supported baud rate specifically for RA device.

9600

115200

500000

1000000

1500000

2000000

4000000

6000000 Note: The maximum value is 1500000 when E2 Lite is selected.

Note: J-Link only supports 9600bps baud rate.

E.g: -connBaudRate 9600

Command Name: -connBaudRate (SWD boot mode)

Prerequisite: None.

Optional.

Description: For SWD boot mode, this command is used to set the SWD connection speed. (in Hz)

Optional: Omitting this command will automatically set the connection speed to a default speed for E2/E2 Lite (1.5MHz) and J-Link (1MHz).

Note: For J-Link, the connection speed is not changeable as it is fixed to 1MHz.

Arguments: SWD connection speed in numeric value of Hz. Some working connection speed for E2/E2 Lite is shown below:

1500000

3000000

6000000 Note: The maximum value is 3000000 when E2 Lite is selected.

Note: J-Link only supports 1000000Hz as the connection speed.

Command Name: -language

Prerequisite: None.

Optional.

Description: The language of the error messages in the console output that the users want to receive in.

Optional: Default language will be English.

Arguments: * english * japanese E.g: -language english / -language JAPANESE

Command Name: -dlmTargetState -action DLM

The target DLM state for the DLM state transition.

Arguments: * SSD - Secure state for debugging. Current DLM state must be CM. * NSECSD - Non-secure state for debugging. Current DLM state must be SSD or OEM(PL2). * DPL - Non-debugging state. Current DLM state must be NSECSD or OEM(PL1). * OEM - Original equipment manufacturer state. Current DLM state must be CM. E.g: -action DLM -dlmTargetState SSD

Command Name: -idauCFS -action BOUNDARY

DLM state must be SSD.

The region size value of Code Secure in kilobytes (kB).

Numeric value in decimal or hexadecimal.

E.g: -action BOUNDARY -idauCFS 256

-action BOUNDARY -idauCFS 0x100

Command Name: -idauCFNSC

Description: The region size value of Code NSC in kilobytes (kB). Numeric value in decimal or hexadecimal.

E.g: -action BOUNDARY -idauCFNSC 4

-action BOUNDARY -idauCFNSC 0x4

Command Name: -idauDFS

Description: The region size value of Data Secure in kilobytes (kB). Numeric value in decimal or hexadecimal.

E.g: -action BOUNDARY -idauDFS 64

-action BOUNDARY -idauDFS 0x40

Command Name: -idauSRAMS

Description: The region size value of SRAM Secure in kilobytes (kB). Numeric value in decimal or hexadecimal.

E.g: -action BOUNDARY -idauSRAMS 64

-action BOUNDARY -idauSRAMS 0x40

Command Name: -idauSRAMNSC

Description: The region size value of SRAM NSC in kilobytes (kB). Numeric value in decimal or hexadecimal.

E.g: -action BOUNDARY -idauSRAMNSC 8

-action BOUNDARY -idauSRAMNSC 0x8

Command Name: -vendorProbe

Prerequisite: Optional. Must be used together with "-emuType other".

Description: Specify the ID of the partner emulator. Omitting this command will default RDPM to automatically select an emulator.

E.g: -emuType other -vendorProbe partneremulator

Command Name: -plTarget

Prerequisite: Optional. Must be used together with "-action PL".

Arguments: * PL1 - Protection Level 1 (Non-secure Software Development) * PL0 - Protection Level 0 (Non-debugging state) E.g: -action PL -plTarget PL1